Daniel Fiifi Tawia Hagan, Ph.D.

Research Scientist

Profile

I am a hydro-climate research scientist with years of experience in hydrological modelling and extreme climate analysis using satellite observations and climate models to assess the impact of climate change on different subsystems of the climate system. I have worked on several international projects aimed at **improving food and water security** in different parts of the world, and I offer consultation aimed at **sustainable developments** for interested stakeholders. I am also the co-host of the Green Living Chats Podcast which is influential in conversations on sustainability with experts from around the world.

Employment History

Senior Research Scientist at Ghent University, Ghent, Belgium

2023 — Present

- Working on identification of drivers for hydrocliamtic extreme events and potential mitigation solutions under Eurpean Research Council's **HEAT** project and Belgium's FWO **CausalHeat** Belgium-China project (Co-PI).
- Early Career Representative of AIMES Land DA working group (https://aimesproject.org/ldawg), NASA Goddard.
- Gues Editor for Frontiers in Environmental Science (https://www.frontiersin.org/research-topics/72027/unravelling-soil-moisture-dynamics-and-their-roles-in-climate-model-sensitivity)
- External collaborator of the Data Assimilation group (DARes) at CISL, National Center for Atmospheric Research (NCAR), USA.
- Invited by the European Space Agency to participat in the ***New Earth Observations Missions Ideas*** workshop at the European Space Research and Technology Centre, Noordwijk, The Netherlands.
- Collaborating with scientists across Europe, North America and Asia to develop causal frameworks to assess changes in the climate system.
- Supervising Graduate students in Belgium, Germany, China, Ghana and India.
- Advisor for the ongoing **IcyAlert** (*Intelligent Climate Early Warning Alert for Arctic Ice-Free Summers*) research project funded by the Novo Nordisk Foundation (https://icyalert.github.io).
- Co-PI for a newly approved Causal Inference for Geoscience by the Singaporean government.
- Reviewer for several high level international journals including npj, Nature, Journal of Hydrology, Remote Sensing, Water, Scientific Data.
- Invited Talk for University of Natural and Energy Resources, Ghana, on Hydro-climate impacts

Freelance Consultant – Sustainability and climate resilience

2019 — Present

- External consultant for the Ghana Meteorological bureaus' hydrological department.
- I offer training/talks on assessing the impact of a warming climate on food and water security.

Details

Ghent, Belgium

danielfhagan@gmail.com

daniel.hagan@ugent.be

Nationality

Ghanaian

Core competences

Data Analysis

Ability to Work in a Team

Climate Modelling

Computer Programming

Causality Analysis

Critical Thinking

Languages

English

French

Fante

Twi

Chinese

Links

ORCID website

Website

Hobbies

Reading, Classical music composition and appreciation, contemporary music arrangement

Co-host of The Green Living Chats Podcast
2019 — Present

- We interview scientsts and personnel from organizations investing in sustainability to address how we can tackle environmental challenges from different perspectives
- We organize programs to help graduate students develop critical thinking to make their research relevant in the society.
- We train young scientists for research in academia and industry.

Research Scientist at Nanjing University of Information Science and Technology (NUIST), Nanjing, China
2020 — 2023

- Developed improvement schemes for soil moisture and LST satellite retrievals from passive microwave observations.
- Contributed to NCAR's soil moisture data assimilation research and used it to study drought forecasts.
- Developed two causality frameworks for flood and drought predictions and land-induced heatwaves events using information flow.
- Worked on Earth Observation Technology-based assessment of the impacts and consequences of climate change.
- Teaching Assistant for a PhD course: Land-climate interactions.

Lecturer at Reading Academy (NUIST) affiliated to the University of Reading (UK), Nanjing
September 2021 — 2023

- Taught the following courses for undergraduates: Linear Algebra, Analyzing Social Data, GIS Programming, Theories and Debates in Human Geography and Research Training for Geography and Environmental Science.

Education

Ph.D., 3Systems Integration & Meteorological Applications, Nanjing University of Information Science & Technology, Nanjing
June 2016 — July 2020

- Graduated with Honours
- Won excellent thesis award
- Contributed to NCAR's data assimilation research as a visiting scholar with the Data Assimilation Lab at the National Center for Atmospheric Research, Boulder, Colorado, USA
- Developed a module for assimilating microwave-based satellite observations as a visiting scholar with the Data Assimilation Lab in Southwest University, Chongqing, China.
- Increased research group's publication records by more than 50% by providing high-quality co-author contributions.
- Taught an undergraduate course: Advance Mathematics.

MSc., Atmospheric Science, Nanjing University of Information Science & Technology, Nanjing
September 2014 — May 2016

Completed after a panel review and a research publication

BSc., Geographic Information Systems, Nanjing University of Information Science & Technology, Nanjing

September 2010 — July 2014

Graduated with Honours

Chinese Language, Nanjing University of Information Science & Technology, Nanjing
September 2009 — July 2010

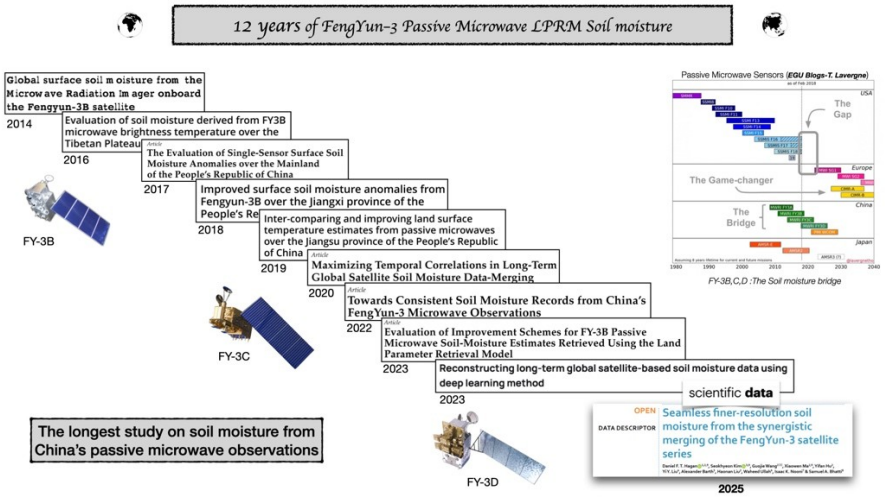
Attained an advanced Chinese language level after a year

Geomatic Engineering, Kwame Nkrumah University of Science and Technology, Kumasi
September 2008 — July 2009

Completed the basic courses and moved to China after a year of study

Research Achievements

- Applied for, received and managed an international research funding under the Belt and Road Initiative of China titled "***Tackling sustainable water resources and food security constraints over the China-Pakistan Economic Corridor in a warming world***."
- Developed Global high resolution daily drought monitoring data for both scientific and operational uses (https://doi.org/10.5281/zenodo.18280747)
- Leading the DARes team at the NCAR(USA) to develop a data assimilation scheme for satellite soil moisture data assimilation using the DART+CLM setup.
- Contributed to several publications on droughts in Africa for historical and future periods.
- Led(since 2015) the longest study on soil moisture from China's passive microwave sensors.



- Developed a long-term merged model dataset for evapotranspiration (1981-2018): https://doi.org/10.5281/zenodo.4595941.
- Developed a time-frequency information flow causality formalism***.
- Improved an existing land surface temperature retrieval algorithm which accounts for open water fraction.
- Developed improvement schemes for the Land Parameter Retrieval Model (LPRM) which accounts for open water fraction and has

improved scattering albedo.

- Obtained two patents for algorithms developed for causality and extreme climate evaluation

Selected Research Publications (80+ SCI publications)

- **Hagan, Daniel Fiifi Tawia**, Wang, G., Kennedy-Asser, A.T. *et al.* Divergent regional responses of soil moisture-air temperature coupling under future climate scenarios. *Nature Communications* **17**, 7311 (2026). https://doi.org/10.1038/s41467-026-74040-w
- Zhou, F., **D. F. T. Hagan**, G. Wang, S. Li, X. Wei, Y. Hu, F. Siddique, and X. Zhang, 2026: Declining Vegetation Response to Soil Moisture and Vapor Pressure Deficit in Inner Mongolian Grasslands. *J. Climate*, **39**, 4617 4634, https://doi.org/10.1175/JCLI-D-25-0597.1
- Emmanuel Yeboah, **Daniel Fiifi Tawia Hagan**, Guojie Wang, Pedro Cabral, Isaac Sarfo, Bright Bediako-Kyeremeh, Chenxia Zhu, Feihong Zhou, Charafa El Rhadiouini, Yifan Hu. Understanding the causal influence of energy balance components on urban heat islands in a mid-latitude city - Chongqing, China. Urban Climate. Volume 65, 2026, 102735, ISSN 2212-0955, https://doi.org/10.1016/j.uclim.2025.102735.
- **Hagan, Daniel Fiifi Tawia**, Kim, S., Wang, G. *et al.* Seamless finer-resolution soil moisture from the synergistic merging of the FengYun-3 satellite series. *Scientific Data* **12**, 983 (2025). https://doi.org/10.1038/s41597-025-05263-7.
- Waheed Ullah, Khameis Alabduoli, Safi Ullah, Sami G. Al-Ghamdi, Khawla Alhebsi, Mansour Almazroui, Mazen E. Assiri, Waqar Azeem, Abdelgadir Abuelgasim, **Hagan, Daniel Fiifi Tawia**, Comparison of 2-m surface temperature data between reanalysis and observations over the Arabian Peninsula, Atmospheric Research, Volume 311, 2024, 107725, ISSN 0169-8095, https://doi.org/10.1016/j.atmosres.2024.107725.
- Shao, Y.; Hagan, D.F.T.; Li, S.; Zhou, F.; Zou, X.; Cabral, P. The Many Shades of the Vegetation–Climate Causality: A Multimodel Causal Appreciation. *Forests* **2024**, *15*, 1430. https://doi.org/10.3390/f15081430
- Nooni, I.K.; Ogou, F.K.; Prempeh, N.A.; Saidou Chaibou, A.A.; **Hagan, Daniel Fiifi Tawia**; Jin, Z.; Lu, J. Analysis of Long-Term Vegetation Trends and Their Climatic Driving Factors in Equatorial Africa. *Forests* **2024**, *15*, 1129. https://doi.org/10.3390/f15071129
- Nooni, I.K.; Ogou, F.K.; **Hagan, Daniel Fiifi Tawia**; Saidou Chaibou, A.A.; Prempeh, N.A.; Nakoty, F.M.; Jin, Z.; Lu, J. The Relationship between Changes in Hydro-Climate Factors and Maize Crop Production in the Equatorial African Region from 1980 to 2021. *Atmosphere* **2024**, *15*, 542. https://doi.org/10.3390/atmos15050542
- **Hagan et al., 2024**, Contrasting Sensitivity of Air Temperature Trends to Surface Soil Temperature Trends Between Climate Models and Reanalyses, **npj Climate and Atmospheric Science.**
- Zou, X.; Wang, G.; **Hagan, Daniel Fiifi Tawia**; Li, S.; Wei, J.; Lu, J.; Qiao, Y.; Zhu, C.; Ullah, W.; Yeboah, E. Precipitation Sensitivity to Soil Moisture Changes in Multiple Global Climate Models. *Atmosphere* **2023**, *14*, 1531. https://doi.org/10.3390/atmos14101531
- Ullah, W., Wang, G., Lou, D., Gao, Z., Zhu, C., Samuel Bhatti, A., **Hagan, Daniel Fiifi Tawia**, Li, S., Jiang, T., Su, B., Karim, A., & Ullah, S. (2023). The Empirical Influence of Tibetan Plateau Spring Soil Moisture on South Asian Monsoon Onset: A Linear Diagnostic Perspective. *Journal of Climate*, *36*(24), 8723-8742. https://doi.org/10.1175/JCLI-D-22-0262.1
- Ullah W, Karim A, Ullah S, Rehman A-U, Bibi T, Wang G, Ullah S, Bhatti AS, Ali G, Abbas A, **Hagan, Daniel Fiifi Tawia**, Nooni IK, Zhu C and Hussain A (2023) An increasing trend in daily monsoon precipitation extreme indices

over Pakistan and its relationship with atmospheric circulations. *Front. Environ. Sci.* 11:1228817. doi: 10.3389/fenvs.2023.1228817

- **Hagan, Daniel Fiifi Tawia**, Dolman Han A. J., Wang G., et al., Contrasting ecosystem constraints on the seasonal terrestrial CO_2 and mean surface air temperature causality projections by the end of the 21st century, Environmental Research Letters, 2022. 10.1088/1748-9326/aca551.
- Mingyue Z.,Guojie W., **Hagan, Daniel Fiifi Tawia**,Waheed U.,Giri K.,Jiao L.,Shijie L., Impacts of vegetation changes on land evapotranspiration in China during 1982-2015, Frontiers in Environmental Science, 2022, 10.3389/fenvs.2022.819277
- Feng D., Wang G., Wei X., Amankwah S.O.Y., Hu Y., Luo Z., **Hagan, Daniel Fiifi Tawia**, Ullah W., Merging and Downscaling Soil Moisture Data From CMIP6 Projections Using Deep Learning Method, Frontiers in Environmental Science, 10, 2022. https://www.frontiersin.org/article/10.3389/fenvs.2022.847475
- Nooni, I.K.; **Hagan, Daniel Fiifi Tawia***; Ullah, W.; Lu, J.; Li, S.; Prempeh, N.A.; Gnitou, G.T.; Lim Kam Sian, K.T.C. *Projections of Drought Characteristics Based on the CNRM-CM6 Model over Africa.* Agriculture 2022, 12, 495. https://doi.org/10.3390/agriculture12040495
- Wang, G.; Ma, X.; **Hagan, Daniel Fiifi Tawia***; Schalie, R.v.d.; Kattel, G.; Ullah, W.; Tao, L.; Miao, L.; Liu, Y. Towards Consistent Soil Moisture Records from China's FengYun-3 Microwave Observations. Remote Sens. 2022, 14, 1225. https://doi.org/10.3390/rs14051225.
- Lu, J.; G. Wang; T. Chen; S. Li; **Hagan, Daniel Fiifi Tawia**; G. Kattel; J. Peng; T. Jiang, and B. Su, 2021: A harmonized global land evaporation dataset from model-based products covering 1980–2017. Earth Syst. Sci. Data, 13, 5879-5898.
- Li, S. G. Wang S. Sun **Hagan, Daniel Fiifi Tawia** T. Chen H. Dolman, and Y. Liu, 2021: Long-term changes in evapotranspiration over China and attribution to climatic drivers during 1980–2010. Journal of Hydrology, 595, 126037
- Nooni, I.K.; **Hagan, Daniel Fiifi Tawia**; Wang, G.; Ullah, W.; Lu, J.; Li, S.; Dzakpasu, M.; Prempeh, N.A.; Lim Kam Sian, K.T.C. *Future Changes in Simulated Evapotranspiration across Continental Africa Based on CMIP6 CNRM-CM6*. Int. J. Environ. Res. Public Health 2021, 18, 6760. https://doi.org/10.3390/ijerph18136760
- Nooni, I.K.; **Hagan, Daniel Fiifi Tawia**; Wang, G.; Ullah, W.; Li, S.; Lu, J.; Bhatti, A.S.; Shi, X.; Lou, D.; Prempeh, N.A.; Lim Kam Sian, K.T.C.; Dzakpasu, M.; Amankwah, S.O.Y.; Zhu, C. Spatiotemporal *Characteristics and Trend Analysis of Two Evapotranspiration-Based Drought Products and Their Mechanisms in Sub-Saharan Africa.* Remote Sens. 2021, 13, 533. https://doi.org/10.3390/rs13030533
- Ullah, W., Guojie, W., Gao, Z., **Hagan, Daniel Fiifi Tawia**, Bhatti, A. S., & Zhua, C. (2021). Observed Linkage between Tibetan Plateau Soil Moisture and South Asian Summer Precipitation and the Possible Mechanism, Journal of Climate, 34(1), 361-377. Retrieved Mar 14, 2021, https://journals.ametsoc.org/view/journals/clim/34/1/JCLI-D-20-0347.1
- **Hagan, Daniel Fiifi Tawia** & Wang, Guojie & Kim, Seokhyeon & Parinussa, Robert & Liu, Yi & Ullah, Waheed & Bhatti, Asher Samuel & Ma, Xiaowen & Jiang, Tong & Su, Buda. Maximizing Temporal Correlations in Long-Term Global Satellite Soil Moisture Data-Merging. Remote Sensing. 2020, 12, 2164. https://www.mdpi.com/2072-4292/12/13/2164
- **Hagan, Daniel Fiifi Tawia** & Guojie Wang & X. San Liang & Han A.J Dolman.

(2019) Time-varying causality formalism based on the Liang-Kleeman information flow for analyzing directed interactions in non-stationary climate systems. Journal of Climate (AMS), Vol 32, No 16. https://doi.org/10.1175/JCLI-D-18-0881.1

- **Hagan, Daniel Fiifi Tawia** & Parinussa, Robert & Wang, Guojie & Clara Draper (2019). An Evaluation of Soil Moisture Anomalies from Global Model-based Datasets over the People's Republic of China. Water, 12(1), 117; https://doi.org/10.3390/w12010117
- **Hagan, Daniel Fiifi Tawia** & Wang, Guojie & Parinussa, Robert & Shi, Shao. (2019) Inter-comparing and improving land surface temperature estimates from passive microwaves over the Jiangsu province of the People's Republic of China. International Journal of Remote Sensing, 40(14): 5563-5584. https://doi.org/10.1080/01431161.2019.1580790
- Liang, Chuanzhuang & Chen, Tiexi & Dolman, Han & Shi, Tingting & Wei, Xueqiong & Xu, Jialu & **Hagan, Daniel Fiifi Tawia**. (2020). "Drying and Wetting Trends and Vegetation Covariations in the Drylands of China." Water 12, no. 4: 933. https://doi.org/10.3390/w12040933
- Yuan, Q. & Wang, G. & Zhu, C. & Lou, D. & **Hagan, Daniel Fiifi Tawia** & Ma, X. & Zhan, M. (2020). Coupling of Soil Moisture and Air Temperature from Multiyear Data During 1980–2013 over China. Atmosphere, 11, 25. https://www.mdpi.com/2073-4433/11/1/25
- Isaac Kwesi Nooni & Guojie Wang & **Daniel Fiifi Tawia Hagan** & Jiao Lu & Waheed Ullah & Shijie Li. (2019). *Evapotranspiration and its Components in the Nile River Basin Based on Long-Term Satellite Assimilation Product*. Water, 11(7): 1400. https://doi.org/10.3390/w11071400
- Parinussa, Robert & Wang, Guojie & Liu, Y. Y & Lou, Dan & **Hagan, Daniel Fiifi Tawia** & Zhan Mingjin & Su, Buda & Jiang, Tong.(2018) Improved surface soil moisture anomalies from Fengyun-3B over the Jiangxi province of the People's Republic of China. International Journal of Remote Sensing, 39(23): 8950-8962. https://doi.org/10.1080/01431161.2018.1500729
- Wang, Guojie & Gong, Tiantian & Lu, Jiao & Lou, Dan & **Hagan, Daniel Fiifi Tawia** & Chen, Tiexi. (2018). On the long-term changes of drought over China (1948–2012) from different methods of potential evapotranspiration estimations. International Journal of Climatology, 38(7): 2954-2966. https://doi.org/10.1002/joc.5475
- Waheed Ullah & Guojie Wang & Gohar Ali & **Daniel Fiifi Tawia Hagan** & Asher Samuel Bhatti & and Dan Lou. (2018). Comparing Multiple Precipitation Products against In-Situ Observations over Different Climate Regions of Pakistan. Remote Sensing, 11(6): 628. https://doi.org/10.3390/rs11060628
- Chen, Tiexi & Zhang, Huixian & Chen, Xing & **Hagan, Daniel Fiifi Tawia** & Wang, Guojie & Gao, Zhiqiu & Shi, Tingting. (2017). Robust drying and wetting trends are found in regions based on Köppen climate classifications over China. Journal of Geophysical Research: Atmospheres, 122(8): 4228-4237. https://doi.org/10.1002/2016JD026168
- Parinussa, Robert & Wang, Guojie & Liu, Yi & **Hagan, Daniel Fiifi Tawia** & Lin, Fenfang & van der Schalie, Robin & de Jeu, Richard. (2017). The Evaluation of Single-Sensor Surface Soil Moisture Anomalies over the Mainland of the People's Republic of China. Remote Sensing. https://doi.org/10.3390/rs9020149
- Wang, Guojie & **Hagan, Daniel Fiifi Tawia** & Lou, Dan & Chen, Tiexi. (2016). Evaluation of soil moisture derived from FY3B microwave brightness temperature over the Tibetan Plateau. Remote Sensing Letters. 7:9, 817-

826. https://doi.org/10.1080/2150704X.2016.1192303

Co-Supervised graduate student thesis (selected)

- *UNRAVELING THE CAUSAL RELATIONSHIP BETWEEN SOIL MOISTURE AND TEMPERATURE*, Doctoral dissertation, 2021, Student: Mr. Samuel Asher Bhatti.
- *THE IMPACT OF SOIL MOISTURE ON THE SUMMER PRECIPITATION AND AIR TEMPERATURE EXTREMES OVER CHINA,* Doctoral dissertation, 2021, Student: Mr. Dan Lou.
- *INVESTIGATING THE MECHANISMS WITHIN THE SOIL MOISTURE-AIR TEMPERATURE INTERACTIONS*, Doctoral dissertation, 2022, Student: Miss Jiao Lu.
- *ASSESSING THE APPLICABILITY AND SPATIO-TEMPORAL CHANGES OF AIRS AND MODIS LAND SURFACE TEMPERATURE*, Master thesis, 2021, Student: Mr. Jiang Liu.
- *GLOBAL SOIL MOISTURE RETRIEVAL AND RECONSTRUCTION FROM THE FENGYUN-3 SATELLITE SERIES*, Master thesis, 2022, Student: Miss Xiaowen Ma.
- *LANDSLIDE DETECTION AND SUSCEPTIBILITY ANALYSIS IN THE HINDU KUSH-KARAKORAM HIMALAYAN REGION USING MACHING/DEEP LEARNING*, Doctoral Dissertation, 2023, Student: Mr. Solomon Obiri Yeboah Amankwah
- *LAND PARAMETER RETRIEVAL AND IMPROVEMENT FROM PASSIVE MICROWAVE SENSORS,* Master Thesis, 2024, Student: Mr. Haonan Liu
- ASSESSING THE SOIL-VEGETATION-ATMOSPHERE FEEDBACK MECHANISMS IN THE CMIP6 RECORDS, Master Thesis, 2024, Student: Miss Xiao Zou
- A MULTI-CLIMATE MODEL ASSESSMENT OF THE EFFECTS OF LAND-SURFACE HYDROTHERMAL CONDITIONS ON AIR TEMPERATURE, Master Thesis, 2024, Student: Miss Yumeng Qiao
- SUBSEASONAL PREDICTION OF HOT SPELLS IN THE IBERIAN PENINSULA USING CAUSALLY-INFORMED MACHINE LEARNING, Master Thesis, 2024, Student: Mr. Cas Decancq
- *SPATIO-TEMPORAL DYNAMICS OF URBAN HEAT ISLANDS IN WESTERN CHONGQING: LAND USE/COVER CHANGE, ENERGY BALANCE MODIFICATION AND POLICY IMPLICATIONS,* Doctoral Dissertation, 2026, Student: Mr. Emmanuel Yeboah
- *CAUSAL INVESTIGATION OF GRASSLAND–CLIMATE INTERACTIONS OVER CHINA*, Doctoral Dissertation, 2026, Student: Mr. Feihong Zhou
- LAND-ATMOSPHERE COUPLING UNDER CLIMATE CHANGE AND HUMAN INFLUENCE: INSIGHTS FROM FULLY COUPLED AND DATA-DRIVEN ANALYSIS, Doctoral Dissertation, 2026, Student: Mr. Yikui Zhang

Courses Taught (Reading Academy of University of Reading, UK and NUIST)

- Theories and Debates in Human Geography, Undergraduate course,

Reading Academy

- Analysing Social Data, BSc course, Reading Academy
- Advanced Mathematics, BSc course, Reading Academy & NUIST
- GIS software development, BSc course, Reading Academy
- Vegetation–climate interactions, PhD course, NUIST
- Land–atmosphere interactions, PhD course, NUIST
- Matlab for Geosciences, MSc course, NUIST